Question 1

Question 1a

Question 1

As shown in Figure 1, a section of a road is made of a concave section and a convex section. A car with a mass of m = 2.0 × 10^3 kg is driving along this road. The radius of curvature (r) for both concave and convex sections is 40 m.

(a) When the car is travelling with a speed of 10 m/s, determine the normal force acting on the car when it is at the lowest point of the concave section.

(15 marks)

$$
\begin{aligned}
v_i &= 10\ ms^{-1} \\
a_c &= \frac{v_i^2}{r} \\
&= \frac{10^2}{40} \\
&= 2.5\ ms^{-2} \\
N &= m(g + a_c) \\
&= 2 \times 10^3 (9.8 + 2.5) \\
&= 24.6\ kN
\end{aligned}
$$

Question 1b

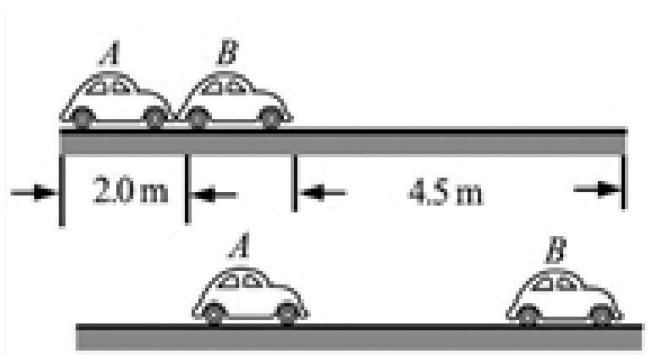
(b) To ensure safety, the car must not leave the ground at any position along the road. Calculate the maximum speed that the car can be travelling at when it reaches the highest point of the convex section without the wheels leaving the ground.

(15 marks)

$$mg = ma_c$$
$$g = a_c$$
$$\frac{v_i^2}{40} = 9.8$$
$$v_i = 19.8\ ms^{-1}$$

Question 2

Question 2a



Question 2

On a winter day, the road was covered in ice, and there was a traffic collision. There were two cars (A and B) on the road. Before the collision, Car A was initially moving towards the right, while Car B was stationary. Figure 2 illustrates the positions of both vehicles at (1) the moment the two cars collide (top), and (2) when both cars come to a complete stop after collision (bottom).

Following the collision, Car B slid forward by 4.5 m, while Car A slid forward by 2.0 m. The given masses of Car A and Car B are 2.0×10^3 kg and 1.5×10^3 kg, respectively. The dynamic friction coefficients between the vehicles and the icy road are 0.10. The collision duration between the two vehicles is extremely brief (can assume elastic collision).

(a) Determine the speed of Car B immediately after the collision. (15 marks)

$$\begin{aligned}
v_B^2 &= u_B^2 + 2as \\
u_B^2 &= v_B^2 - 2as \\
u_B &= \sqrt{-2as} \\
&= \sqrt{2\mu g s} \\
&= \sqrt{2 \times 0.1 \times 9.8 \times 4.5} \\
&= 2.97\ ms^{-1}
\end{aligned}$$

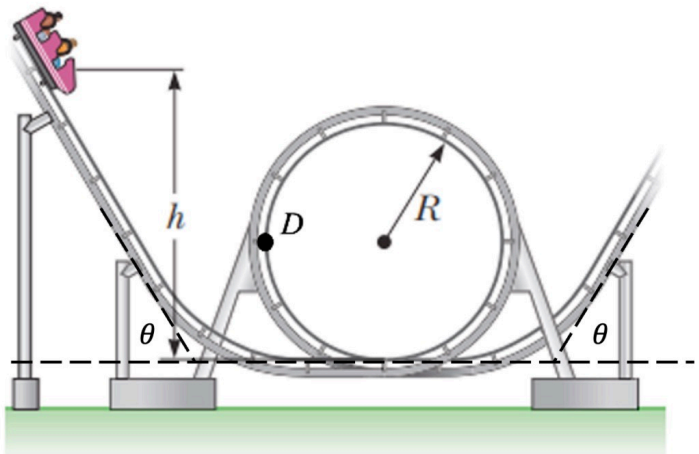
Question 2b

(b) Determine the speed of Car A immediately before the collision. (15 marks)

$$\begin{aligned}
v_A^2 &= u_A^2 + 2as \\
u_A^2 &= v_A^2 - 2as \\
u_A &= \sqrt{-2as} \\
&= \sqrt{2\mu g s} \\
&= \sqrt{2 \times 0.1 \times 9.8 \times 2} \\
&= 1.98\ ms^{-1} \\
m_{A0} v_{A0} &= m_{A1} v_{A1} + m_{B1} v_{B1} \\
v_{A0} &= \frac{2 \times 10^3 \cdot 1.98 + 1.5 \times 10^3 \cdot 2.97}{2 \times 10^3} \\
&= 4.21\ ms^{-1}
\end{aligned}$$

Question 3

Question 3a



Question 3

Figure 3 shows a section of a roller coaster ride. The car is initially at rest on a slope with angle θ. The height of the car is h = 100 m. Later, the brake is released and the car begins to travel along the track. The total mass of the car is 100 kg. The vertical circular loop of the track has a fixed radius R = 20 m.

(a) Assuming no friction, determine the car's speed when it enters the circular loop.

(8 marks)

$$
\begin{aligned}
E_{GP} &= E_k \\
mgh &= \frac{1}{2}mv^2 \\
gh &= \frac{1}{2}v^2 \\
9.8 \times 100 &= \frac{1}{2} \times v^2 \\
v &= 44.3\ ms^{-1}
\end{aligned}
$$

Question 3b

(b) Determine the support force acting on the car by the track when it reaches position *D*, which is at the same height as the center of the circular loop.

(8 marks)

The support force that the track exerts on the car is the normal force.

$$\begin{aligned} N &= F_c \\ &= \frac{mv^2}{r} \\ &= \frac{100 \times 44.3^2}{20} \\ &= 9.81\ kN \end{aligned}$$

Question 3c

(c) Point out where the car and passengers will experience the maximum support force and calculate this force.

(8 marks)

The maximum normal force occurs at the bottom of the loop.

$$\begin{aligned} N &= m(g + a_c) \\ &= 100(9.8 + \frac{v^2}{r}) \\ &= 100(9.8 + \frac{44.3^2}{20}) \\ &= 10.8\ kN \end{aligned}$$

Question 3d

(d) The maximum force that the car and its passengers can withstand is 25000 N, determine the maximum speed that it can be travelling at without causing injury.

(8 marks)

$$\begin{aligned} N &\leq m(g + \frac{v^2}{r}) \\ 25000 &\leq 100(9.8 + \frac{v^2}{20}) \\ v &\leq 69.3\ ms^{-1} \end{aligned}$$

Question 3e

(e) When the car exits the circular loop, it begins to climb a slope with the same elevation angle θ. Taken into account of energy conservation, describe what physical forces prevent it from reaching a height of 100 m on the right-hand side of the figure in real life.

(8 marks)

The car is able to reach the same height as the initial on the right-hand side if all energy is conserved. However, in real life, energy can be lost due to the following physical forces:

- Friction between the wheels and the track
- Air resistance

The lost energy is transferred to heat and sound.